
Testing Cone Alignment of Perceptual Diminishing Returns with Public Color-Dissimilarity Data

Reproducible research report

Abstract

Does the compression of large perceived color differences act separately on cone channels? We test this prediction using historical human dissimilarities, a gray-axis replication, and a nested family of additive, globally compressed, and per-axis compressed distances. Cone models face five standard non-cone systems: opponent (both sum and mean conventions), linear sRGB, CIE RGB, CIELAB, and LCh, all equally parameterized. On pair-disjoint cross-validation, the cone model improves mean squared error over global compression by 0.01166 (95% training-bootstrap CI [0.01050, 0.01403]), but its advantage over the best control is -0.00789 ([-0.00963, -0.00623]); positive values favor cones. The opponent basis wins the aggregate comparison. Polygon-path curvature controls and direct spectral cone conversion preserve the absence of a cone advantage. A controlled two-channel comparison also favors an opponent alternative. Synthetic recovery shows effect-dependent discrimination and weak recovery of individual compression functions, despite good prediction in some regimes. Conservative uncertainty over held-out pairs includes a small cone advantage, so the result is not population-wide equivalence or a neural-site identification. The contribution is a reproducible, selection-aware test of cone specificity and its resolution limits, beyond demonstrating chromatic diminishing returns or anisotropy.

1 Introduction

A distance along a geodesic is additive: if B lies between A and C on that geodesic, then $D(A, C) = D(A, B) + D(B, C)$. Perceptual diminishing returns (DR) violate this equality. [Bujack et al. \[2022\]](#) demonstrated a concave difference-scaling function on the neutral axis, conditional on that axis being a perceptual geodesic. [Stark et al. \[2025\]](#) extended the evidence to two chromatic paths and showed that the same global correction does not transfer between them. Their discussion proposed a decomposition into three cone functions and an outer function, while recognizing that two paths could not test it. A separate follow-up formalized color attributes in non-Riemannian geometry and investigated the relation between perceptual and induced-metric paths [[Bujack et al., 2025](#)]. Neither follow-up establishes cone alignment.

The distinction between *direction-dependent compression* and *cone-specific compression* matters. If three stable cone functions generated compression throughout color space, a small set of measurements could replace a direction-by-direction atlas. It would also make a functional prediction about processing order: compression aligned with individual L , M , S signals is compatible with a mechanism before channel combination, whereas compression aligned with $L - M$ or $S - (L + M)/2$ is compatible with a later combination stage. Behavioral fit alone cannot locate a neural mechanism, even with perfect model discrimination. It can, however, reject the prediction that cones are the privileged axes within a stated family. If a standard non-cone system matches or improves the prediction, the claimed cone-specific economy and localization have not been established.

We test that prediction using the supplied Helm and Ekman dissimilarities, which cover more directions than the two chromatic lines but introduce different limitations. Hue loci are curved; rated

differences are not forced-choice probabilities; and published coordinates do not uniquely determine modern cone excitations. We therefore separate response scales, explicitly control sampled-locus curvature, and test recovery at the actual stimulus directions. Our three questions are: (a) do per-cone inner functions improve held-out prediction over one outer scaling; (b) do they improve it over the *best* of the matched fixed control panel; and (c) are separate cone functions or a coarser channel split identifiable?

The contributions are the controlled alignment test, its selection-aware uncertainty, and a measured account of what these public data resolve. The result is qualified and negative for cone specificity: an apparent benefit from per-axis compression does not yield a cone advantage, even at the tested coarse resolution. This is neither another demonstration that chromatic DR exists nor a claim that the entire visual system lacks cone-organized nonlinearities.

2 Methodology

2.1 A declared cone proxy and fixed comparison coordinates

Let $X \in \mathbb{R}^{n \times 3}$ contain the supplied CIE 1931 color-matching functions and $F \in \mathbb{R}^{n \times 3}$ the supplied CIE 2006 energy cone fundamentals at common wavelengths. We derive

$$T^\top = \arg \min_B \|XB - F\|_F^2, \quad \mathbf{c} = \text{diag}(T\mathbf{W})^{-1}T\mathbf{x}_{XYZ}. \quad (1)$$

The stimulus path is $xyY \rightarrow XYZ \rightarrow \text{LMS}$; for gray it is $L^*a^*b^* \rightarrow XYZ \rightarrow \text{LMS}$. Y is divided by 100. Illuminant $\mathbf{C}$, $\mathbf{W} = (0.31006/0.31616, 1, 0.37378/0.31616)$, is the documented Helm white and an explicitly nominal analysis white for Ekman. Gray uses D65, $(0.95047, 1, 1.08883)$. Thus a reference white maps to $(1, 1, 1)$ by von Kries normalization. We set the tabulated missing long-wavelength S values to zero.

Equation 1 is a CIE-2006-derived *linear proxy*, not an exact change of observer. The two supplied standards do not share precisely the same spectral span [Stockman, 2019]. Negative projected components are retained and audited, not silently clipped. For Ekman’s nominal monochromatic lights we also evaluate the provided cone fundamentals directly, using energy proportional to $Y/\bar{y}(\lambda)$, with the same white normalization. This is a prespecified sensitivity analysis. Without surface reflectance spectra, an analogous exact Helm reconstruction is unavailable. All biological interpretations are conditional on this colorimetry.

The non-cone panel uses fixed coordinates: opponent $(L - M, S - (L + M), L + M)$, plus the white-balanced convention $(L - M, S - (L + M)/2, (L + M)/2)$; linear sRGB; CIE RGB; standard CIELAB; and L^*, C^*, h with $C^* = \sqrt{a^{*2} + b^{*2}}$, $h = \text{atan2}(b^*, a^*)$. Linear RGB uses the disclosed XYZ matrices without clipping, gamma encoding, or fitted adaptation. For LCh, $|\Delta h|$ is the shorter angular difference in radians. Thus the last two controls use their own nonlinear coordinates, not a stimulus-dependent linearization. No rotation is estimated. Appendix A gives every transform.

For each fixed basis and dataset we divide each axis difference by its root mean square over the *stimulus design*’s unique pairs. This response-independent diagonal scaling gives all axes comparable numerical units without rotating any basis. A constant axis remains zero. Consequently, shape comparisons are explicitly conditional on the white and design normalization; functions fitted to different datasets are not claimed to be a universal common scale.

2.2 A nested concave family with a fixed gauge

Write $x_k = |\Delta c_k|/s_k$ for the normalized absolute differences. Define

$$h_p(t) = \begin{cases} \tau\{(1 + t/\tau)^p - 1\}/p, & p > 0, \\ \tau \log(1 + t/\tau), & p = 0, \end{cases} \quad \tau = 0.1, \quad 0 \leq p \leq 1, \quad (2)$$

$$I_k(x) = x - (1 - q_{k1})(x - 0.25)_+ - (q_{k1} - q_{k1}q_{k2})(x - 0.75)_+, \quad 0 \leq q_{kj} \leq 1. \quad (3)$$

The inner slopes are $1, q_{k1}, q_{k1}q_{k2}$. All functions are nondecreasing and concave. We impose $h(0) = I_k(0) = 0$, $h'(0) = I'_k(0) = 1$, and $\sum_k w_k = 1$, $w_k > 0$. An observation amplitude supplies the output unit. These constraints fix the multiplicative and offset gauges; restricting the outer function to one shape parameter limits, but does not eliminate, finite-design trade-offs between inner and outer curvature.

$$\begin{aligned}
\text{M0 : } D &= \sum_k w_k x_k, \\
\text{M1 : } D &= h_p(\sum_k w_k x_k), \\
\text{M2 : } D &= h_p(\sum_k w_k I_k(x_k)) \quad \text{on cone coordinates,} \\
\text{M3}_b : D &= h_p(\sum_k w_k I_k(x_k^{(b)})) \quad \text{on each fixed control basis } b.
\end{aligned} \tag{4}$$

Setting every q to one gives M1; additionally setting $p = 1$ gives M0. M0, M1, and M2 have respectively two, three, and nine structural parameters in three dimensions. Each M3 also has nine. The Gaussian amplitude and residual SD bring the totals to four, five, and eleven. A constant coordinate has lower effective identification, not extra fitted flexibility; in particular Ekman has constant L^* . We retain the same nominal functional family in all controls.

M0 is weighted city-block distance, not a Riemannian L_2 metric. M1 already allows linear directional weighting; the M2–M1 question concerns additional *nonlinear* axis dependence. A better M2 fit establishes neither a universal violation of Riemannian geometry nor a unique neural implementation.

For coarse resolution we define $A = (L + M)/2$. C1 applies only h_p to weighted differences in (A, S) ; C2 adds the two inner functions. Q2 is the matched inner-plus-outer model on $(L - M, S - A)$. C2 and Q2 each have six structural parameters, or eight including Gaussian nuisance parameters. (A, S) already mixes a derived and a cone signal: supporting it would not establish separate pre-combination L and M compression.

2.3 Separate observation models and held-out estimands

For pair ratings, $y_{ijr} \sim \mathcal{N}(aD_{ij}, \sigma^2)$, with $a > 0$ and no intercept, so zero distance predicts zero dissimilarity. Helm responses are divided by a fixed factor of 20; Ekman values are used as supplied. Each dataset has its own latent parameters, amplitude, and variance. Minimizing squared error profiles a analytically; σ is estimated from training residuals over individual observed ratings. We report out-of-fold MSE and Gaussian log likelihood, with the latter including its normalization constant. Held-out likelihood uses only training estimates of a and σ .

Our primary chromatic statistic is the equally weighted average of the two datasets’ mean squared errors in those declared response units. It is a descriptive meta-average, not a pooled likelihood or an assertion of equal perceptual units. Per-dataset comparisons are indispensable because these units and designs differ. Define

$$\delta_{21} = \text{MSE}(\text{M1}) - \text{MSE}(\text{M2}), \quad \delta_{23} = \min_b \text{MSE}(\text{M3}_b) - \text{MSE}(\text{M2}). \tag{5}$$

Positive values favor cones. We recompute the minimum inside each paired bootstrap replicate, including after meta-averaging; the reported alignment interval never conditions on a point-selected winner.

For the gray experiment the response probability is

$$\Pr(R = 1) = \Phi\{a[f(|g(L_s^*) - g(L_{t2}^*)|) - f(|g(L_s^*) - g(L_{t1}^*)|)]\}. \tag{6}$$

Following the reference pipeline, training trials first estimate a monotone suprathreshold map g , then hold it fixed while comparing linear f to h_p . We use a monotone cubic Hermite interpolant at five fixed $L^*/100$ knots, positive increments, and endpoints zero and one. A prespecified free-log bridge, $f_k(d) = \log(1 + kd)/k$, permits magnitude comparison without the fixed outer family’s lower-bound restriction. We also run direct L^* and relative-luminance anchors, whose compression can include ordinary stimulus-to-strength nonuniformity.

The independent Wiebel MLDS observations use a probit comparison of the two adjacent intervals of a ten-level monotone scale, with fixed endpoints and fitted noise. Each viewing context is fitted separately. Three cone components on a gray axis cannot identify a three-direction decomposition; alignment is evaluated on the chromatic sets. MLDS supplies suprathreshold interval scales; its additive observation model *cannot independently prove DR*. We do not transfer these scales between displays or pool their likelihoods with gray trials or pair ratings.

2.4 Cross-validation, uncertainty, and fitting

All primary fits use five fixed folds. Chromatic folds are disjoint by unordered pair, gray folds by unique triple and stratified by reference level, and MLDS folds by participant. Every competing

model receives identical folds. There is no held-out hyperparameter tuning. The timestamped analysis plan fixes the families, knots, normalization, comparison, recovery, and confound analyses before the fitting scripts were written. A later formula audit identified that the requested opponent convention used the sum, while our plan used its mean. A timestamped addendum, written before fitting the added convention, retains both in the final six-member control panel and repeats nearest-control recovery; no earlier fit is discarded.

We refit all models in 200 training-only bootstrap replicates. Within each chromatic fold we resample training pairs and, for Helm, training observer columns; gray resamples training triads and their empirical Bernoulli responses; MLDS resamples training participants. Test responses remain unchanged. The displayed percentile 95% bootstrap CIs therefore quantify training-sample uncertainty *conditional on the supplied test panels*; they are not population CIs over new colors. Curves use the average of the five training-fold functions, with the analogous bootstrap ensemble. Full-data curves are explicitly descriptive where shown.

To expose uncertainty due to the small held-out pair panels, we also report paired Student intervals from out-of-fold pair losses, separately from training bootstrap uncertainty. For the selected control we construct simultaneous intervals for all six gaps at individual coverage $1 - 0.05/6$ (Bonferroni 95% family coverage) and take the minimum of their lower and upper endpoints. This supplementary approximation treats pairs as sampling units; shared stimuli and overlapping training sets preclude an unconditional population guarantee. For the meta-average, variances add after multiplying each dataset by one half.

L-BFGS-B uses analytic derivatives for pair models, bounded weight logits in $[-6, 6]$, and the shape bounds above. Point cross-validation uses four starts, descriptive full fits six, and bootstrap refits two, one initialized from the corresponding *training-fold* fit. Synthetic fits use three starts and four for full recovery. Starts are chosen by training loss alone. Convergence, boundaries, objectives, seeds, folds, and bootstrap indices are saved.

2.5 Curvature and identifiability controls

For gray we order each triple along intensity and retain it only if $(\|A - B\| + \|B - C\|) / \|A - C\| - 1 \leq 0.01$ in normalized cone coordinates. This numerical collinearity check does not by itself prove a perceptual geodesic.

For each chromatic set we form a closed polygon in the provided stimulus sequence. Two genuinely additive geometry baselines use a fitted diagonal Euclidean chord metric and its shortest polygon-path metric. Their outer-compressed counterparts test the incremental value of a scalar compression. Separately, we rerun the complete M0–M3 panel and C1/C2/Q2 using accumulated absolute axis increments along the shorter path in a *fixed*, RMS-normalized cone Euclidean polygon. The path is chosen from coordinates, never from responses. All comparison bases use that same path; LCh edges use circular hue increments. Appendix D distinguishes this controlled path construction from the fitted Euclidean baseline. Ekman’s closing endpoint edge is unobserved. These analyses assess aliasing by the sampled locus; they cannot certify unknown perceptual geodesics.

We diagnose signed and absolute cone-difference designs with SVD and correlations. Synthetic Gaussian recovery uses exactly the observed pairs, replicate count, folds, and an empirical residual SD. For each design and both chord/path modes, we run 100 datasets per generator (M2, opponent M3, C2, Q2), with both fixed unequal shapes and empirical-fit shapes. No synthetic observation enters a human-data result. We measure parameter/curve recovery, correct held-out model ordering, and the frequency of a paired-loss interval excluding zero in the correct direction. Wilson 95% CIs quantify Monte Carlo uncertainty. The same coarse diagnostics test C2 against C1 and against Q2; they prevent a bare, untested assertion of non-identifiability.

3 Experimental Setup

Public data and provenance. The gray release contains 91,896 forced-choice responses to 320 triads [Bujack et al., 2022]. The two supplied release filenames contain identical values and are used once. The public columns have no participant identifiers, so participant-disjoint validation is impossible; triple-disjoint validation avoids repeating a test triple in training. The `lan1/color` reference implementation and release are supplied under BSD-3-Clause.

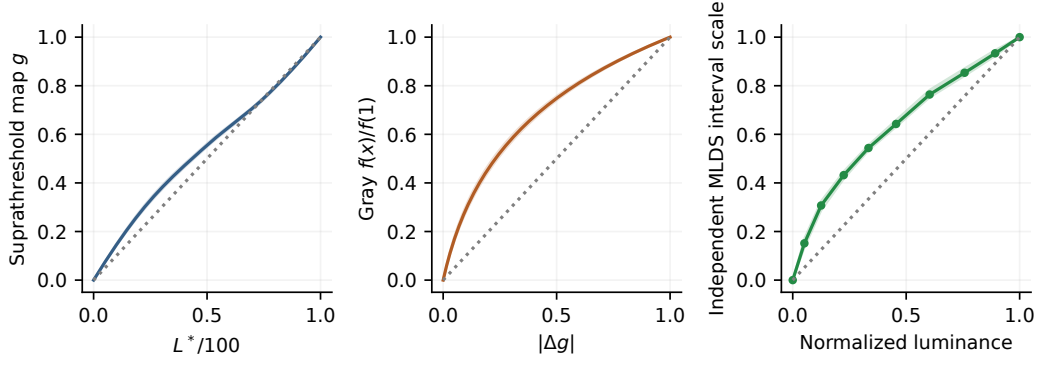


Figure 1: Independent empirical anchors. Left: the fitted gray suprathreshold map. Middle: the free-log difference scaling, normalized at one for display. Right: the independent plain-context MLDS interval scale. Lines are descriptive full-data fits; shading is the central 95% training-bootstrap distribution of fold-averaged curves. MLDS validates interval-scale estimation but does not test DR. Dashed diagonals show linear mappings.

Helm’s ten surface-color tiles provide 45 unordered pairs [Helm, 1964]. We use the color-normal observers, averaging N6a/N6b within observer N6 before combining the ten observers, for 450 observer–pair values. We exclude the color-deficient matrices because their sensor population does not match the standard cone model. Ekman provides 91 aggregate dissimilarities among 14 spectral-color labels [Ekman, 1954]. We use the supplied $1 - \text{similarity}$ matrix, without applying that conversion twice. Both matrices are distributed by `smacof` under GPL-3 [Mair et al., 2022]; the supplied factual colorimetry is attributed to the original measurements and CVRL.

The 2017 lightness data contain 60,000 MLDS responses from ten participants in five contexts [Wiebel et al., 2017]. They are supplied from OSF under CC BY 4.0. Physical luminances are read from the matching files; matching *responses* are not treated as dissimilarities. The cone fundamentals and XYZ functions are supplied from CVRL, whose citation and redistribution terms are not a permissive software license. We use no additional human data and collect none. Appendix B records sources, exclusions, and limitations.

Colorimetric verification. The projection in Eq. 1 has relative spectral RMS errors of 2.25%, 2.31%, and 13.32% for L, M, and S. These are deterministic calibration residuals, not statistical estimates. Both whites map exactly to $(1, 1, 1)$; CIELAB round trips are checked. At the independent 555-nm reference the projected S value is negative even though the tabulated S fundamental is positive (Appendix A). Two Ekman components are negative under the projection, while none are under direct spectral conversion. This verification reveals a material limitation of deriving modern cones from historical XYZ, rather than validating an exact physiological transformation. The direct-cone sensitivity is consequently essential. Ekman’s supplied common Y is nominal: equal subjective brightness does not imply measured equal photometric luminance. Conclusions concern that provided embedding.

Empirical gray anchor before extension. Our own out-of-fold gray fit improves negative log likelihood by 0.00606 nats/trial (95% CI [0.00550, 0.00698]); the supplementary held-out-triad interval is [0.00428, 0.00784]. The fitted fixed-family exponent is 0.000 ([0.000, 0.081]), at the logarithmic boundary. The free-log bridge estimates $k = 10.15$ ([8.86, 11.83]). Its compression index

$$R_f = \frac{f(0.30)}{2f(0.15)} \quad (7)$$

is 0.755 ([0.743, 0.767]), versus the published illustration $2.06/(2 \times 1.36) = 0.757$ [Bujack et al., 2022]. This scale-free magnitude comparison uses intervals on our endpoint-normalized suprathreshold scale, not an assertion that the two fitted maps or noise units coincide. All 320 triples pass the collinearity restriction, so its before/after score is identical. Direct L^* and luminance fits, including their poorer predictive scores, are retained in Appendix G.

Table 1: Primary results. Chromatic scores are the equally weighted MSE of Helm/20 and Ekman; lower scores are better. Gains are comparator loss minus target loss, so positive favors the first named model. All losses/gains are multiplied by 10^3 ; the gray compression ratio is unscaled. Intervals are paired, training-only bootstrap 95% CIs conditional on the observed test panels. The best M3 is reselected in each bootstrap. C2/Q2 is the controlled coarse-resolution test, not a three-cone identification.

Quantity	Estimate	95% CI	Comparison	Notes
Gray DR gain	6.061	[5.497, 6.976]	M1 vs. M0	nats/trial
Gray compression ratio	0.755	[0.743, 0.767]	2022: 0.757	log bridge
M0: additive	105.669	[104.801, 116.992]	held-out MSE	
M1: outer DR	30.409	[30.228, 34.353]	held-out MSE	
M2: cones	18.750	[18.229, 21.504]	held-out MSE	
M3: opponent mean	11.174	[11.040, 13.935]	held-out MSE	matched M3
M3: sRGB	26.519	[25.276, 32.478]	held-out MSE	matched M3
M3: CIE RGB	18.442	[17.981, 22.522]	held-out MSE	matched M3
M3: CIELAB	28.111	[27.831, 32.626]	held-out MSE	matched M3
M3: LCh	13.176	[12.910, 15.051]	held-out MSE	matched M3
M3: opponent sum	10.862	[10.877, 13.688]	held-out MSE	matched M3
Direction-dependent gain	11.659	[10.496, 14.029]	M2 vs. M1	positive
Cone-alignment gain	-7.888	[-9.632, -6.232]	M2 vs. best M3	opponent wins
Coarse organization gain	52.805	[51.183, 55.623]	C2 vs. C1	positive
Coarse alignment gain	-18.021	[-20.462, -16.289]	C2 vs. Q2	Q2 wins
After curvature control	-7.436	[-8.736, -6.213]	M2 vs. best M3	opponent wins

Independent interval-scale check. The plain-context MLDS fit yields held-out negative log likelihood 0.430 ([0.425, 0.450]) and improves over a physical-luminance linear scale by 0.078 nats/trial ([0.063, 0.081]). The other contexts and noise estimates appear in Appendix G. These nonlinear stimulus-to-strength maps reinforce the need to distinguish g from the difference scaling f ; they are not additional cone-alignment observations.

4 Experimental Results

4.1 Direction dependence does not establish cone alignment

Table 1 gives the required nested and matched-panel comparison. The aggregate improvement of M2 over M1 is 0.01166 ([0.01050, 0.01403]), but it is driven by Ekman: 0.02429 ([0.02169, 0.02819]), compared with -0.00097 ([-0.00219, 0.00193]) for Helm. Thus inner cone functions help on the spectral design, whereas additional nonlinear axis dependence is not established on the surface-color design. Neither an averaged gain nor a large training improvement should be read as a uniform effect across color space.

The sum-convention opponent model has the lowest aggregate held-out error. M2’s selection-aware advantage over the panel is -0.00789 ([-0.00963, -0.00623]), so the required cone-specificity criterion fails. Per dataset, the winning control is LCh for Helm and opponent for Ekman; their selection-aware cone gaps are -0.00234 ([-0.00530, -0.00046]) and -0.01476 ([-0.01792, -0.01308]). The stronger perceptual-coordinate controls have been evaluated directly, with no local-frame approximation. Figure 2 shows every required model, and Appendix C reports their separate scores and likelihoods.

The uncertainty estimand matters. A simultaneous interval based on held-out pair variability for the aggregate selected-panel gap is [-0.01604, 0.00027], which includes zero and a small positive value. It is wider than the fixed-test training-bootstrap interval. We therefore report a robust *failure to establish a cone advantage on these panels*, and a conditional predictive preference for opponent coordinates, rather than claiming population-wide rejection or statistical equivalence of all cone mechanisms. Nor do we infer that a preference for a named opponent coordinate system uniquely localizes compression after neural aggregation.

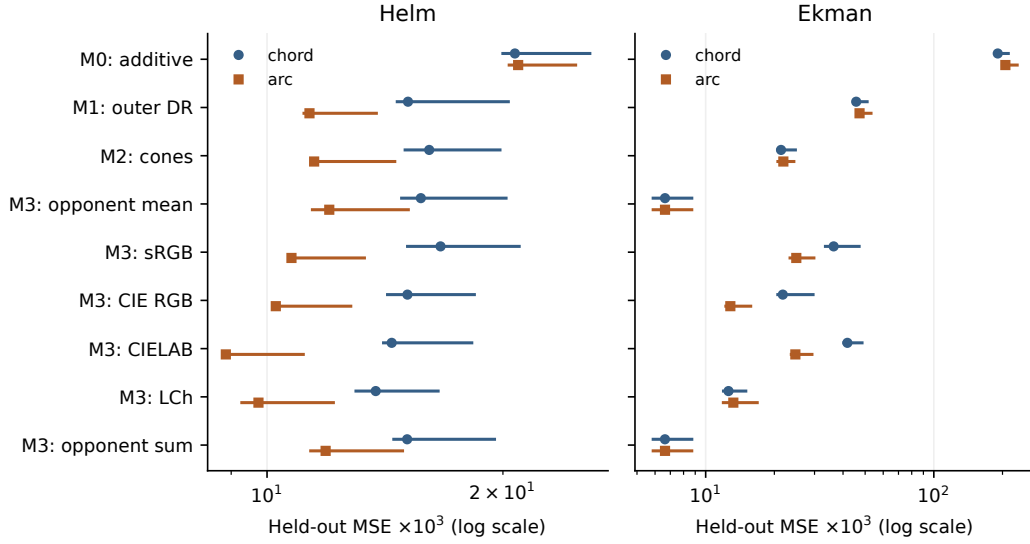


Figure 2: The complete fixed-basis panel on identical pair-disjoint folds. Markers are out-of-fold MSE, with training-bootstrap 95% intervals. Circles use endpoint differences; squares use accumulated differences on a common fixed polygon path. The horizontal axes are logarithmic and have different ranges. A cone advantage must survive comparison with the whole panel.

4.2 What is identifiable at full and coarse resolution?

Figure 3 shows the fitted inner contributions and their unit-slope shapes. The fitting family has not been forced to collapse to M1: at normalized difference one, the M–S shape contrast is -0.550 ($[-0.678, -0.355]$) for Helm and 0.089 ($[0.062, 0.206]$) for Ekman. However, the L–M contrasts are 0.550 ($[-0.031, 0.629]$) and 0.011 ($[-0.098, 0.239]$), both including zero. These are contrasts of inner shapes, distinct from axis weights. They do not establish three separate biological functions. In particular, the Ekman full fit concentrates weight near the M direction, and bootstrap variation redistributes weight and curvature (Appendix E).

The design diagnostics do not support a blanket claim that L and M always move together. In Helm the signed L–M correlation is negative and the normalized design has full rank; in nominal equal-Y Ekman it has rank two. Taking absolute values makes the latter design full rank, so signed rank deficiency alone cannot establish non-identifiability of a nonlinear distance model. We use recovery to test the stronger claim (Figure 4; full diagnostics in Appendix E). Individual inner-slope recovery is broad even in the fixed unequal-shape simulations; several central recovery intervals reach both boundaries. Predictive basis discrimination can nevertheless be substantially more reliable than recovery of all function parameters.

The coarse (A, S) model C2 improves on its outer-only baseline by 0.05281 ($[0.05118, 0.05562]$), but loses to matched two-axis Q2 by -0.01802 ($[-0.02046, -0.01629]$). Again, Ekman carries the organization gain: 0.10580 ($[0.10398, 0.11142]$); Helm’s gain is -0.00019 ($[-0.00280, 0.00209]$). The coarse null is thus a tested prediction, not an inference from an ill-conditioned design alone. It supports no privileged (A, S) organization and no separate L/M localization.

Table 2 conditions power on empirical-fit alternatives and the observed designs. On Ekman, the C2–C1 organization test has detection probability 1.00 (Wilson CI $[0.96, 1.00]$). Q2-generated coarse data are discriminated in Q2’s favor with probability 1.00 ($[0.96, 1.00]$). For the full-dimensional opponent alternative, empirical sum-convention data are discriminated with probability 0.95 ($[0.89, 0.98]$); the mean-convention result is 0.92 ($[0.85, 0.96]$). These diagnostics earn the negative predictive result at substantial fitted effects. They do not make every cone-vs-opponent null informative: fixed moderate alternatives and cone-generated data can have much lower significance frequencies, especially after the path control. All such cases are included in Appendix F.

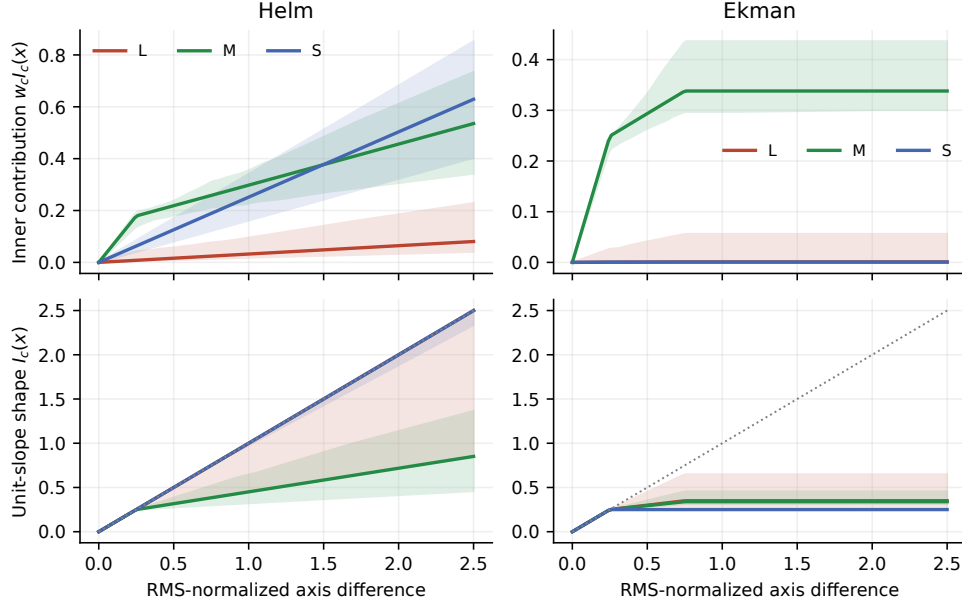


Figure 3: Actual fitted inner contributions $f_c = w_c I_c$ (top) and their unit-initial-slope shapes I_c (bottom), averaged across the five training folds. Shading is the pointwise 95% interval across 200 training-bootstrap ensembles. The normalization is fixed within each design. Distinct-looking curves can coexist with uncertain weight allocation and an unresolved L–M contrast; they are not estimates of receptor response laws.

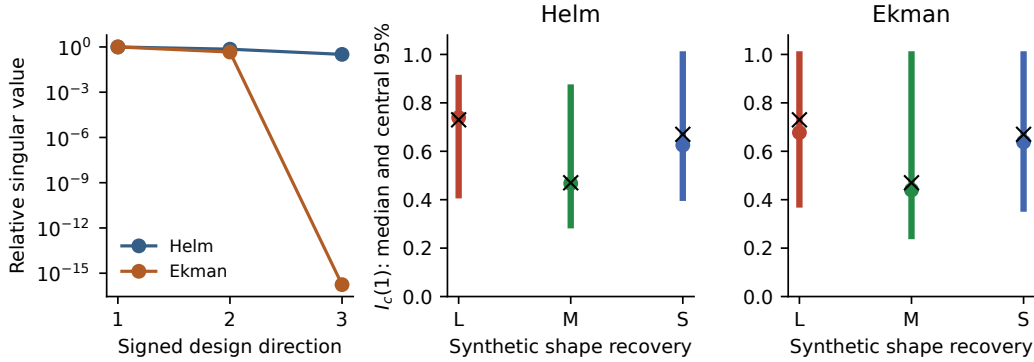


Figure 4: Resolution diagnostics. Left: singular values of the signed RMS-normalized cone-difference design, relative to the largest. Ekman’s third direction is at numerical precision. Right: recovery of $I_c(1)$ from 100 M2-generated datasets using fixed unequal shapes and empirical noise; dots are medians, vertical lines central 95% recovery ranges, and crosses the generating values. These are synthetic recovery ranges, not human-data confidence intervals.

4.3 Curvature and conversion sensitivities

Replacing endpoint differences with accumulated polygon-path differences leaves the aggregate direction-dependent gain positive, 0.01260 ([0.01159, 0.01495]), while the selected-panel cone advantage remains negative, -0.00744 ([-0.00874, -0.00621]). The coarse-model gap against Q2 also remains negative, -0.01690 ([-0.01906, -0.01548]). These conclusions do not require treating all colors on a hue loop as lying on one straight geodesic. The simultaneous held-out-pair interval for the path-controlled cone gap remains wide, [-0.01602, 0.00103].

Table 2: Synthetic discrimination at empirical-fit alternatives, with 100 repetitions per row and Wilson 95% CIs. “Correct sign” compares M2 to mean-convention opponent M3, or C2 to Q2, in the generating model’s direction. “Detected alignment” additionally requires the paired held-out interval to exclude zero. The last column detects M2 over M1 or C2 over C1, regardless of generator; it is not necessarily a test of the generating model. These simulations use common residual noise within each competitor pair and are not human observations. Exact sum-convention results are reported in Appendix F.

Design	Generator	Correct sign	Detected alignment	Detected organization
Helm	M2	1.00 [0.96, 1.00]	0.63 [0.53, 0.72]	0.34 [0.25, 0.44]
Helm	M3	0.99 [0.95, 1.00]	0.40 [0.31, 0.50]	0.01 [0.00, 0.05]
Helm	C2	1.00 [0.96, 1.00]	0.80 [0.71, 0.87]	0.67 [0.57, 0.75]
Helm	Q2	1.00 [0.96, 1.00]	0.98 [0.93, 0.99]	0.00 [0.00, 0.04]
Ekman	M2	0.91 [0.84, 0.95]	0.27 [0.19, 0.36]	1.00 [0.96, 1.00]
Ekman	M3	1.00 [0.96, 1.00]	0.92 [0.85, 0.96]	0.98 [0.93, 0.99]
Ekman	C2	1.00 [0.96, 1.00]	0.61 [0.51, 0.70]	1.00 [0.96, 1.00]
Ekman	Q2	1.00 [0.96, 1.00]	1.00 [0.96, 1.00]	1.00 [0.96, 1.00]

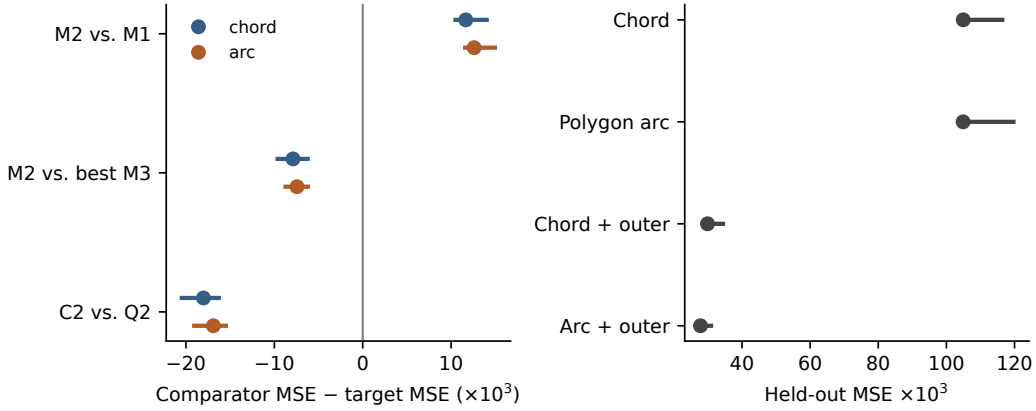


Figure 5: Before/after sampled-locus curvature control. Left: cone and coarse-model gains for endpoint versus polygon-path features. Right: additive diagonal Euclidean chord and polygon-arc baselines, with and without a scalar outer compression. Intervals use paired training bootstraps. The scalar compression improves these baselines, but no sampled polygon is certified as the perceptual geodesic.

The explicit additive chord and arc baselines in Figure 5 show that simply replacing a straight distance by a length along the provided polygon is insufficient to account for the rated differences in this family. Adding scalar compression improves both. Nonetheless, a nonlinear perceptual embedding or the response link can also produce concavity along these loci. The chromatic observations therefore remain DR/curvature-aliased with respect to unrestricted perceptual geometry. They support the reported *controlled prediction* comparisons, not a new proof of chromatic non-Riemannian geometry.

Direct spectral cone conversion changes numerical scores but preserves the Ekman findings: the M2–M1 gain becomes 0.01265 ([0.01126, 0.01495]), while the selected-panel cone gain is -0.01064 ([-0.01271, -0.00983]), still against the opponent winner. This matters because the primary projection can produce nonphysical small negative components. Complete results for both conversions, both locus treatments, every control, and the additive geometry baselines are supplied in the appendices and machine-readable logs.

5 Analysis and Failure Modes

Three distinct verdicts. *Direction-dependent nonlinearity:* M2 improves over M1 on the nominal spectral design, but the corresponding Helm result is unresolved. *Cone alignment:* M2 fails the

declared best-control criterion in the primary comparison, after the polygon-path control, and under direct spectral conversion. The trained models prefer opponent coordinates for Ekman; uncertainty over a broader population of pairs is less decisive. *Granularity*: the data support neither three separately identified cone functions nor a privileged coarse (A, S) decomposition. At coarse resolution they can discriminate substantial alternatives, and the matched opponent alternative performs better. These statements are compatible: parameter recovery, model discrimination, and neural localization are different inferential tasks.

Why the null is earned, and where it is not. The conclusion does not rely on a generic assertion that L and M are correlated. We ran the full panel, quantified signed and absolute design rank, checked individual function recovery, and tested the coarse model against both its nested baseline and a matched alternative. The calibrated simulations show high sensitivity to some fitted alternatives, including an opponent generator close to the cone basis. They also show low sensitivity to more modest effects and to some cone-generated regimes. Thus our negative answer is bounded by the tested family, effect sizes, observation noise, and stimulus coordinates. A non-cone match defeats cone *specificity* here; it does not imply the visual system contains no cone-wise compression.

The main failures are scientific, not numerical. All recorded chromatic bootstrap fits and synthetic recovery fits report optimizer convergence, but many parameters lie on bounds. A zero later inner slope, negligible axis weight, or $p = 0$ can provide good prediction while offering poor parameter identification. In Helm, the absence of a held-out M2 improvement prevents treating in-sample distinct shapes as evidence for full cone organization. In Ekman, exact equal- Y planarity, nominal brightness calibration, and reduced effective dimension in CIELAB/LCh constrain interpretation. Its aggregate matrix cannot support a participant-level uncertainty analysis. The gray release likewise cannot recover participant clusters. The shared-pair and shared-stimulus structure limits both bootstrap and closed-form population interpretations.

Model and gauge limitations. The city-block family is a deliberate fixed aggregation choice. Some of the control advantage could reflect its axis-dependent geometry as well as nonlinear compression, because rotating an L_1 distance changes even its linear form. Accordingly, an M3 win does not isolate a post-receptoral *nonlinearity* without comparing additional within-basis additive/outer-only controls. The fitted coarse comparison tests privileged alignment, not a unique processing stage. Our fixed knots, bounded logits, and single outer family leave model misspecification possible. We report the complete planned sensitivity set and the disclosed coefficient-convention audit rather than select a normalization or basis that favors cones. The origin and derivative gauge removes formal scale freedom, but sparse finite differences leave curvature trades and do not identify receptor laws.

Implication for compact prediction. A transferable three-function cone model would justify extrapolating from a few channel measurements to new directions. These independent datasets do not establish that transfer: their response scales, backgrounds, stimuli, and design normalizations differ, and we fit them separately. Nor does the successful coarse discrimination establish an achromatic/chromatic biological split. The practical outcome is a falsifiable benchmark and an explicit resolution boundary: a future whole-space model must outperform strong fixed perceptual and opponent controls on new directions, with reliable colorimetry and enough near-geodesic samples to separate compression from locus geometry. This report supplies the computational test, not those missing human measurements.

6 Related Work

The 2022 neutral-axis study introduced the two-stage suprathreshold/difference-scaling analysis used as our replication anchor [Bujack et al., 2022]. Its result motivates the present question but does not identify chromatic channel axes. *Now in Color* establishes compression on two chromatic paths and non-transfer of a global correction [Stark et al., 2025]; our novel estimand is instead whether cone axes outperform a matched fixed-basis panel. The geometry follow-up formalizes perceptual attributes and addresses induced-length versus perceptual paths, including empirical response-bias controls [Bujack et al., 2025]. It does not estimate a per-cone decomposition. These distinctions prevent relabeling existing anisotropy or geometry results as an alignment discovery.

The historical Helm and Ekman matrices motivated multidimensional accounts of color dissimilarity [Helm, 1964, Ekman, 1954]; our analysis conditions on their supplied physical coordinates rather than estimating an unrestricted embedding. Their availability in `smacof` supports reproducible reuse but does not recover lost trial-level metadata [Mair et al., 2022]. MLDS estimates suprathreshold interval scales from comparative judgments [Maloney and Yang, 2003, Knoblauch and Maloney, 2008]; Wiebel et al. [2017] connects those scales with appearance matches. We use the supplied MLDS observations to verify scale estimation while keeping its additive assumptions separate from the DR test. Modern cone standards motivate physiologically meaningful coordinates, while differences between standard observers require the conversion audit and spectral sensitivity reported here [Stockman, 2019].

7 Conclusion

Public color-dissimilarity data permit a substantive alignment test, but they do not establish a privileged cone decomposition of diminishing returns. The gray replication reproduces concavity at a comparable magnitude. On the spectral-color design, per-axis compression improves held-out prediction; the cone basis nevertheless fails to beat the best standard non-cone control. The conclusion persists under the sampled-path and spectral-conversion sensitivities, and the coarse comparison favors its matched opponent alternative at effects the simulations can detect. Individual cone-function recovery remains weak, and broader pair uncertainty prevents a universal negative claim. The contribution beyond prior chromatic DR work is the controlled cone-specificity test and an empirical account of its identifiable resolution, without claiming pre- or post-combination neural localization.

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A Conversion, Normalization, and Fixed Bases

The common spectral grid is 390–830 nm at 1-nm spacing. There are 215 missing entries in the S column at the long-wavelength end; they are set to zero before the least-squares projection. No spectral data are downloaded. The resulting transform, in the supplied fundamental units, is

$$T = \begin{pmatrix} 0.26827402 & 0.84661357 & -0.03489198 \\ -0.38594472 & 1.16487727 & 0.10285125 \\ 0.02124671 & -0.02455204 & 0.53416092 \end{pmatrix}. \quad (8)$$

All stimulus coordinates, normalization factors, checks, and the full-precision matrix are in `logs/conversion.json` and `logs/coordinates_*.csv`. The least-squares objective uses equal spectral sampling weights, not an illuminant spectrum; reference-white normalization is applied afterward. Thus Helm’s Illuminant C chromaticities determine its XYZ normalization without reconstructing its unknown reflectance spectra.

For a known chromatic reference at 555 nm, the projected unnormalized fundamental vector is (0.983783, 0.967846, −0.010601), whereas the supplied tabulated vector is (0.965733, 0.956583, 0.001203). This discrepancy is explicitly retained as a failed exact-conversion check. Normalized C and D65 whites are (1, 1, 1) by construction. The absence of an exact XYZ-to-modern-LMS transform follows from the spectral residual, not from an optimizer failure. Under the direct Ekman calculation, the relative RMS differences from the projection are 17.15%, 11.00%, and 4.04% across L, M, and S. These descriptive conversion discrepancies have no sampling CIs.

The fixed XYZ-to-linear-sRGB matrix is [International Color Consortium, 2015]

$$B_{sRGB} = \begin{pmatrix} 3.2406255 & -1.537208 & -0.4986286 \\ -0.9689307 & 1.8757561 & 0.0415175 \\ 0.0557101 & -0.2040211 & 1.0569959 \end{pmatrix}. \quad (9)$$

We define CIE RGB using

$$\mathbf{x}_{XYZ} = B_{CIE}\mathbf{x}_{RGB}, \quad B_{CIE} = \begin{pmatrix} 2.768892 & 1.751748 & 1.130160 \\ 1 & 4.5907 & 0.0601 \\ 0 & 0.056508 & 5.594292 \end{pmatrix}. \quad (10)$$

These transformations use the same XYZ vectors as the cone projection. They are equivalently fixed linear maps of the primary cone differences because T is invertible. For the direct-spectral sensitivity, the original XYZ-based non-cone coordinates remain fixed, while the cone and opponent coordinates change. This avoids silently redefining the RGB controls to follow the alternate cones.

For CIELAB, with $\epsilon = (6/29)^3$ and $u(t) = t^{1/3}$ above ϵ , $u(t) = t/[3(6/29)^2] + 4/29$ below it,

$$L^* = 116u(Y/Y_n) - 16, \quad (11)$$

$$a^* = 500\{u(X/X_n) - u(Y/Y_n)\}, \quad (12)$$

$$b^* = 200\{u(Y/Y_n) - u(Z/Z_n)\}. \quad (13)$$

The white is C for both chromatic sets. LCh uses these exact coordinates and the circular convention $\Delta h = \text{wrap}_{[-\pi, \pi)}(h_i - h_j)$ before taking the absolute value. It is the CIELAB cylindrical system, not CIELUV HCL and not a chroma-weighted hue chord. All conventions are fixed in advance and remain the same in bootstrap and recovery fits.

The design scaling is $s_k = \sqrt{N^{-1} \sum_{i < j} (\Delta z_{ij,k})^2}$. For LCh the circular differences enter this formula. A numerically constant coordinate receives $s_k = 1$ and retains zero differences; hence its nominal parameters are unidentifiable. Table 3 records all factors for the primary coordinates. They depend only on the stimulus tables, not on responses or fold assignment. Diagonal scaling preserves basis directions but does not equalize their correlation condition numbers; fully whitening them would rotate away the hypothesis being tested.

Table 3: RMS axis-difference divisors in coordinate order. Values of one on a constant L^* axis are inactive-axis conventions, not estimated variation.

Design	Basis	Divisors
Helm	cone	0.016300, 0.032118, 0.196413
Helm	opp	0.045596, 0.191307, 0.011352
Helm	opp_sum	0.045596, 0.186753, 0.022705
Helm	srgb	0.208442, 0.067928, 0.244605
Helm	ciergb	0.027605, 0.006336, 0.041041
Helm	lab	0.610708, 25.745848, 29.180709
Helm	lch	0.610708, 5.210469, 1.944193
Helm	coarse	0.011352, 0.196413
Helm	qcoarse	0.045596, 0.191307
Ekman	cone	0.225954, 0.513577, 11.439125
Ekman	opp	0.450787, 11.115977, 0.326506
Ekman	opp_sum	0.450787, 10.793031, 0.653012
Ekman	srgb	2.163424, 1.905731, 14.244348
Ekman	ciergb	0.230886, 0.050144, 2.385830
Ekman	lab	1.000000, 316.397031, 279.263123
Ekman	lch	1.000000, 266.941216, 1.756050
Ekman	coarse	0.326506, 11.439125
Ekman	qcoarse	0.450787, 11.115977

B Data Provenance and Units

Table 4: Only the supplied public human data enter the analysis. Licenses refer to the supplied distribution, not ownership of factual color coordinates.

Source	Observations retained	Provenance / terms
Bujack et al. (2022)	Gray triad responses; duplicate CSV copies used once	https://github.com/lanl/color ; BSD-3-Clause
Wiebel et al. (2017)	MLDS trials; physical luminance columns from matching files	https://osf.io/58jz3/ ; CC BY 4.0
Helm (1964)	Normal observers only; N6 repeats averaged	Supplied smacof matrices, GPL-3; supplied Table-I colorimetry attribution
Ekman (1954)	Supplied aggregate $1 - s$ dissimilarities	Supplied smacof matrix, GPL-3; spectral coordinates and nominal common Y supplied
CVRL	Supplied LMS and XYZ spectral functions	https://cvrl.org ; citation/redistribution terms, no permissive software license asserted

Helm’s response units are divided by 20 for a fixed numerical scale, without claiming a physical equivalence to Ekman’s bounded dissimilarities. Changing this constant would change the aggregate weighting but not either per-dataset comparison. The surface-color table uses authoritative x, y, Y rather than the internally inconsistent historical X/Z transcription noted in the supplied provenance. We do not fill the missing Munsell hue label by inference. Ekman’s wavelength table is treated as the supplied nominal spectral specification; equal brightness, missing spectral bandwidths, and nominal luminance limit physiological conclusions. We do not reconstruct undocumented stimulus spectra or participant judgments.

The MLDS matching files contain rounded display labels for reflectance but exact stimulus-level luminance columns. We sort the unique reflectance levels and map the trial indices to this order, without recomputing physical luminance from the rounded labels. Reverse-order triads are put into ascending order and the binary response is reversed correspondingly. Contexts remain separate. Their interval scales are not imported into the 2022 gray map because the displays and contexts differ.

C Full Cross-Validated Scores

Tables 5 and 6 include every required control and coarse model for both colorimetric specifications. All intervals in these tables are conditional training-bootstrap 95% CIs. Direct Ekman uses the same human responses and folds as the primary Ekman analysis and therefore is a sensitivity analysis, not an independent replication. Log likelihood is reported separately in Table 7; likelihoods from different response types are never summed.

Table 5: Endpoint-difference MSE $\times 10^3$ with 95% CIs.

Model	Helm	Ekman	Direct Ekman
M0: additive	20.716 [19.998, 25.819]	190.623 [188.849, 211.994]	122.331 [119.244, 139.501]
M1: outer DR	15.131 [14.662, 20.319]	45.686 [44.821, 51.009]	28.237 [27.287, 31.956]
M2: cones	16.105 [15.005, 19.827]	21.395 [20.714, 24.728]	15.583 [15.125, 17.706]
M3: opponent mean	15.714 [14.852, 20.184]	6.635 [5.887, 8.692]	4.943 [4.423, 5.949]
M3: sRGB	16.650 [15.112, 20.978]	36.388 [33.494, 47.053]	36.388 [33.487, 49.096]
M3: CIE RGB	15.107 [14.250, 18.389]	21.778 [20.714, 29.558]	21.778 [20.246, 27.797]
M3: CIELAB	14.424 [14.083, 18.251]	41.799 [40.361, 48.452]	41.799 [40.308, 47.796]
M3: LCh	13.763 [12.987, 16.532]	12.589 [11.970, 14.991]	12.589 [12.065, 14.774]
M3: opponent sum	15.094 [14.514, 19.511]	6.630 [5.888, 8.693]	4.940 [4.417, 5.950]
C1: A, S outer	18.404 [18.571, 22.116]	147.096 [145.256, 154.268]	98.114 [97.180, 102.607]
C2: A, S inner	18.596 [17.700, 22.979]	41.294 [40.215, 44.283]	32.420 [32.057, 34.320]
Q2: opponent inner	16.439 [15.513, 21.142]	7.408 [6.618, 8.668]	4.986 [4.641, 6.253]

Table 6: Accumulated-path MSE $\times 10^3$ with 95% CIs.

Model	Helm	Ekman	Direct Ekman
M0: additive	20.915 [20.372, 24.763]	205.848 [199.932, 231.927]	127.632 [125.813, 151.737]
M1: outer DR	11.328 [11.148, 13.787]	47.251 [46.790, 53.048]	29.180 [28.511, 33.029]
M2: cones	11.484 [11.427, 14.555]	21.901 [20.731, 24.350]	15.575 [15.028, 18.287]
M3: opponent mean	12.007 [11.426, 15.146]	6.634 [5.887, 8.691]	4.945 [4.417, 5.949]
M3: sRGB	10.744 [10.716, 13.313]	24.958 [23.436, 29.809]	24.958 [23.740, 29.906]
M3: CIE RGB	10.261 [10.399, 12.790]	12.827 [12.262, 15.752]	12.827 [12.387, 15.719]
M3: CIELAB	8.859 [8.968, 11.123]	24.723 [23.736, 29.228]	24.723 [23.927, 29.717]
M3: LCh	9.751 [9.286, 12.152]	13.219 [11.955, 16.823]	13.219 [12.183, 16.643]
M3: opponent sum	11.879 [11.370, 14.890]	6.635 [5.888, 8.690]	4.944 [4.419, 5.949]
C1: A, S outer	13.303 [12.948, 16.678]	147.828 [145.968, 154.904]	99.268 [98.308, 103.783]
C2: A, S inner	12.609 [12.373, 16.100]	41.293 [40.215, 44.282]	32.420 [32.057, 34.320]
Q2: opponent inner	12.735 [12.120, 15.728]	7.373 [6.601, 8.604]	4.976 [4.638, 6.246]

Table 7: Mean held-out Gaussian log likelihood per rating (larger is better), with training-bootstrap CIs; SD is across five fold MSE values, multiplied by 10^3 . Positive continuous-density log likelihoods are valid. The observer-averaged Helm mean fit uses the individual observer residuals to estimate variance.

Design	Model	Log likelihood	95% CI	Fold SD
Helm	M0: additive	0.492	[0.208, 0.528]	9.240
Helm	M1: outer DR	0.655	[0.303, 0.679]	5.920
Helm	M2: cones	0.618	[0.255, 0.662]	5.714
Helm	M3: opponent mean	0.639	[0.274, 0.651]	4.122
Helm	M3: sRGB	0.587	[0.198, 0.661]	6.724
Helm	M3: CIE RGB	0.648	[0.273, 0.680]	5.706
Helm	M3: CIELAB	0.678	[0.342, 0.663]	5.284
Helm	M3: LCh	0.683	[0.243, 0.702]	6.117
Helm	M3: opponent sum	0.663	[0.297, 0.659]	3.480
Ekman	M0: additive	-0.592	[-0.654, -0.589]	23.295
Ekman	M1: outer DR	0.120	[0.041, 0.131]	8.841
Ekman	M2: cones	0.454	[0.139, 0.508]	12.704
Ekman	M3: opponent mean	1.070	[0.788, 1.120]	2.008
Ekman	M3: sRGB	0.199	[-0.137, 0.257]	16.629
Ekman	M3: CIE RGB	0.456	[0.028, 0.479]	8.784
Ekman	M3: CIELAB	0.137	[-0.043, 0.181]	20.629
Ekman	M3: LCh	0.741	[0.535, 0.781]	5.173
Ekman	M3: opponent sum	1.070	[0.788, 1.119]	2.013

D Curvature Baselines and Before/After Comparisons

Let e_{rk} denote absolute normalized coordinate increments on polygon edge r . The additive diagonal Euclidean chord is $a\sqrt{\sum_k w_k x_{ij,k}^2}$. Its polygon-arc counterpart is

$$a \min \left\{ \sum_{r \in P_{ij}^+} \sqrt{\sum_k w_k e_{rk}^2}, \sum_{r \in P_{ij}^-} \sqrt{\sum_k w_k e_{rk}^2} \right\}. \quad (14)$$

Thus its selected arc is shortest under its fitted diagonal metric. Both have two independent weights, amplitude, and Gaussian SD. Their compressed counterparts add only p . They are supplementary L2 geometry baselines, distinct from the nested city-block models.

For the full fixed-basis panel, in contrast, we choose P_{ij} once using the unweighted Euclidean length of the RMS-normalized cone polygon and replace $x_{ij,k}$ by $\sum_{r \in P_{ij}} e_{rk}$. A common, response-independent path prevents basis-specific path selection from becoming hidden flexibility. Directions and polygon routes are saved in the conversion log. The spectral path is closed by an endpoint edge, so no claim about the missing nonspectral path follows. The differences between the two arc constructions are intentional and disclosed.

Table 8: Additive-metric curvature baselines and their scalar-compressed versions. $\text{MSE} \times 10^3$, 95% training-bootstrap CIs. These fits are shared across the endpoint/path panel displays, not independently refitted alternatives selected after inspection.

Model	Helm	Ekman	Direct Ekman
Additive chord	18.132 [17.447, 23.468]	191.729 [188.889, 214.498]	121.056 [117.484, 141.789]
Additive arc	18.842 [18.104, 22.371]	190.958 [189.671, 220.754]	131.952 [125.580, 152.910]
Compressed chord	14.441 [14.192, 20.268]	45.288 [44.845, 51.933]	28.497 [27.348, 32.330]
Compressed arc	10.323 [10.322, 12.549]	45.299 [44.645, 50.668]	28.924 [28.102, 32.271]

Table 9: Per-dataset gains before and after the path control, multiplied by 10^3 . Positive favors the first model. Selected-panel intervals reselect the winner per resample.

Design	Features	Comparison	Gain	95% CI
Helm	chord	M2 vs. M1	-0.974	[-2.185, 1.928]
Helm	chord	M2 vs. best M3	-2.342	[-5.305, -0.465]
Helm	chord	C2 vs. C1	-0.192	[-2.795, 2.092]
Helm	chord	C2 vs. Q2	-2.157	[-5.291, 1.170]
Ekman	chord	M2 vs. M1	24.292	[21.685, 28.195]
Ekman	chord	M2 vs. best M3	-14.765	[-17.916, -13.078]
Ekman	chord	C2 vs. C1	105.802	[103.983, 111.415]
Ekman	chord	C2 vs. Q2	-33.885	[-36.810, -32.158]
Helm	arc	M2 vs. M1	-0.156	[-1.757, 0.665]
Helm	arc	M2 vs. best M3	-2.625	[-4.458, -1.372]
Helm	arc	C2 vs. C1	0.694	[-1.168, 1.971]
Helm	arc	C2 vs. Q2	0.126	[-2.552, 1.772]
Ekman	arc	M2 vs. M1	25.350	[24.093, 30.292]
Ekman	arc	M2 vs. best M3	-15.266	[-17.045, -13.304]
Ekman	arc	C2 vs. C1	106.535	[104.675, 112.325]
Ekman	arc	C2 vs. Q2	-33.920	[-36.821, -32.197]

E Design and Parameter Identifiability

Table 10: Deterministic diagnostics of the cone-difference design. Columns give signed rank, signed L–M correlation, absolute L–M correlation, and condition numbers for signed and absolute RMS-normalized differences. These design quantities have no sampling CIs.

Design	Rank	r_{LM}	$r_{ L M }$	$\kappa(\Delta c)$	$\kappa(\Delta c)$
helm	3	-0.703	0.409	3.105	3.755
ekman	2	0.628	0.725	5.73×10^{15}	5.316
ekman direct	3	0.383	0.461	13.011	4.225

Constant XYZ luminance gives an exact linear constraint on the primary Ekman cone proxy. The absolute-value design can nevertheless span three dimensions, and inner functions can distinguish rotated models on that plane. It would be incorrect to infer observational equivalence solely from signed rank two. Conversely, a full-rank absolute design does not establish recovery of the separate weights, inner slopes, and outer curvature. The simulations and empirical intervals test this remaining issue directly.

Table 11: M2 full-fit weights and inner slopes, with conditional 95% intervals from the training-bootstrap ensemble averaged over folds. Boundary estimates and asymmetric intervals are retained. $s_1 = q_1$, $s_2 = q_1 q_2$. The intervals describe the training-ensemble estimator, so they need not be centered on the descriptive full fit.

Design	Cone	w	95% CI	s_1	95% CI	s_2	95% CI
Helm	L	0.038	[0.018, 0.172]	1.000	[0.301, 1.000]	1.000	[0.253, 1.000]
Helm	M	0.770	[0.553, 0.778]	0.141	[0.097, 0.541]	0.141	[0.091, 0.484]
Helm	S	0.192	[0.161, 0.341]	1.000	[1.000, 1.000]	1.000	[0.906, 1.000]
Ekman	L	0.000	[0.000, 0.107]	0.000	[0.000, 0.800]	0.000	[0.000, 0.000]
Ekman	M	0.998	[0.891, 0.998]	0.170	[0.125, 0.411]	0.000	[0.000, 0.010]
Ekman	S	0.002	[0.002, 0.002]	0.000	[0.000, 0.000]	0.000	[0.000, 0.000]

Table 12: Contrasts of unit-initial-slope inner functions at normalized difference one, averaged over the five training folds. These are shape contrasts, not weighted contribution contrasts.

Design	Contrast	Estimate	95% CI
Helm	L – M	0.550	[-0.031, 0.629]
Helm	L – S	0.000	[-0.531, 0.000]
Helm	M – S	-0.550	[-0.678, -0.355]
Ekman	L – M	0.011	[-0.098, 0.239]
Ekman	L – S	0.100	[0.000, 0.400]
Ekman	M – S	0.089	[0.062, 0.206]

Table 13: Synthetic M2 parameter recovery for the fixed unequal-shape generator on endpoint features. Intervals are central 95% recovery ranges over 100 simulations, not empirical-data CIs. Wide intervals expose parameter ambiguity despite some successful basis discrimination.

Design	Parameter	Truth	Median recovery	Recovery range
Helm	$q_{L,1}$	0.80	0.768	[0.225, 1.000]
Helm	$q_{L,2}$	0.40	0.432	[0.208, 1.000]
Helm	$q_{M,1}$	0.40	0.399	[0.073, 1.000]
Helm	$q_{M,2}$	0.20	0.193	[0.000, 1.000]
Helm	$q_{S,1}$	0.60	0.531	[0.213, 1.000]
Helm	$q_{S,2}$	0.80	0.946	[0.339, 1.000]
Ekman	$q_{L,1}$	0.80	0.680	[0.194, 1.000]
Ekman	$q_{L,2}$	0.40	0.482	[0.080, 1.000]
Ekman	$q_{M,1}$	0.40	0.323	[0.000, 1.000]
Ekman	$q_{M,2}$	0.20	0.404	[0.000, 1.000]
Ekman	$q_{S,1}$	0.60	0.608	[0.155, 1.000]
Ekman	$q_{S,2}$	0.80	0.713	[0.000, 1.000]

F All Synthetic Discrimination Checks

The fixed full-dimensional generator has uniform weights, $p = 0.6$, and $(q_{L1}, q_{L2}) = (0.8, 0.4)$, $(q_{M1}, q_{M2}) = (0.4, 0.2)$, $(q_{S1}, q_{S2}) = (0.6, 0.8)$; the opponent generator uses the same shapes on its own axes. The fixed coarse generator uses the first two pairs with equal weights and the same p . These are specified synthetic alternatives, not fitted human parameters or targets for the empirical optimizer. Preset amplitudes match the RMS of observed mean responses. Empirical generators instead use the corresponding full-data fitted functions and amplitudes.

Within a three-axis generator pair, the noise SD is the full M2 residual SD; within a coarse pair it is the C2 residual SD. We generate ten independent Gaussian measurements per Helm pair and one per Ekman pair. This isolates model discrimination under a declared noise model. It does not recover unknown real observer covariance, and the Gaussian simulations can extend outside rating bounds. The simulation code and saved generator parameters make that approximation explicit. Complete uncertainty is recorded for both correct-sign probability and significance frequency.

For each simulated dataset, the same five folds are refitted. Let ℓ_i be a paired held-out squared-error difference. The detection event is that $\bar{\ell} \pm t_{.975, N-1} s_{\ell} / \sqrt{N}$ excludes zero in the generator’s direction. As in the empirical supplementary interval, this is a pair-unit diagnostic, not an exact repeated-CV test. The independent simulation repetitions permit direct estimation of how often that declared rule succeeds. We also report correct-sign probability, which does not depend on an internal standard-error threshold.

Table 14: All fixed-generator discrimination results. Columns follow Table 2, with Wilson 95% CIs. Low detection frequencies are retained, especially for the nearest opponent alternative in the Ekman preset regime.

Design	Features	Gen.	Correct sign	Detected alignment	Detected organization
Helm	chord	M2	1.00 [0.96, 1.00]	0.80 [0.71, 0.87]	0.12 [0.07, 0.20]
Helm	chord	M3	1.00 [0.96, 1.00]	0.64 [0.54, 0.73]	0.00 [0.00, 0.04]
Helm	chord	C2	1.00 [0.96, 1.00]	1.00 [0.96, 1.00]	0.20 [0.13, 0.29]
Helm	chord	Q2	1.00 [0.96, 1.00]	1.00 [0.96, 1.00]	0.00 [0.00, 0.04]
Helm	arc	M2	0.89 [0.81, 0.94]	0.23 [0.16, 0.32]	0.02 [0.01, 0.07]
Helm	arc	M3	1.00 [0.96, 1.00]	1.00 [0.96, 1.00]	0.00 [0.00, 0.04]
Helm	arc	C2	1.00 [0.96, 1.00]	1.00 [0.96, 1.00]	0.12 [0.07, 0.20]
Helm	arc	Q2	1.00 [0.96, 1.00]	1.00 [0.96, 1.00]	0.00 [0.00, 0.04]
Ekman	chord	M2	1.00 [0.96, 1.00]	0.67 [0.57, 0.75]	0.01 [0.00, 0.05]
Ekman	chord	M3	0.58 [0.48, 0.67]	0.02 [0.01, 0.07]	0.04 [0.02, 0.10]
Ekman	chord	C2	0.63 [0.53, 0.72]	0.06 [0.03, 0.12]	0.03 [0.01, 0.08]
Ekman	chord	Q2	1.00 [0.96, 1.00]	0.76 [0.67, 0.83]	0.98 [0.93, 0.99]
Ekman	arc	M2	0.86 [0.78, 0.91]	0.13 [0.08, 0.21]	0.02 [0.01, 0.07]
Ekman	arc	M3	0.54 [0.44, 0.63]	0.00 [0.00, 0.04]	0.04 [0.02, 0.10]
Ekman	arc	C2	0.72 [0.63, 0.80]	0.02 [0.01, 0.07]	0.02 [0.01, 0.07]
Ekman	arc	Q2	1.00 [0.96, 1.00]	0.86 [0.78, 0.91]	1.00 [0.96, 1.00]

Table 15: All empirical-generator discrimination results, including the path-controlled regime. No generator is chosen as the sole calibration of a universal null.

Design	Features	Gen.	Correct sign	Detected alignment	Detected organization
Helm	chord	M2	1.00 [0.96, 1.00]	0.63 [0.53, 0.72]	0.34 [0.25, 0.44]
Helm	chord	M3	0.99 [0.95, 1.00]	0.40 [0.31, 0.50]	0.01 [0.00, 0.05]
Helm	chord	C2	1.00 [0.96, 1.00]	0.80 [0.71, 0.87]	0.67 [0.57, 0.75]
Helm	chord	Q2	1.00 [0.96, 1.00]	0.98 [0.93, 0.99]	0.00 [0.00, 0.04]
Helm	arc	M2	0.94 [0.88, 0.97]	0.28 [0.20, 0.37]	0.14 [0.09, 0.22]
Helm	arc	M3	0.99 [0.95, 1.00]	0.23 [0.16, 0.32]	0.00 [0.00, 0.04]
Helm	arc	C2	1.00 [0.96, 1.00]	1.00 [0.96, 1.00]	0.67 [0.57, 0.75]
Helm	arc	Q2	1.00 [0.96, 1.00]	1.00 [0.96, 1.00]	0.00 [0.00, 0.04]
Ekman	chord	M2	0.91 [0.84, 0.95]	0.27 [0.19, 0.36]	1.00 [0.96, 1.00]
Ekman	chord	M3	1.00 [0.96, 1.00]	0.92 [0.85, 0.96]	0.98 [0.93, 0.99]
Ekman	chord	C2	1.00 [0.96, 1.00]	0.61 [0.51, 0.70]	1.00 [0.96, 1.00]
Ekman	chord	Q2	1.00 [0.96, 1.00]	1.00 [0.96, 1.00]	1.00 [0.96, 1.00]
Ekman	arc	M2	0.98 [0.93, 0.99]	0.22 [0.15, 0.31]	1.00 [0.96, 1.00]
Ekman	arc	M3	1.00 [0.96, 1.00]	0.94 [0.88, 0.97]	1.00 [0.96, 1.00]
Ekman	arc	C2	0.99 [0.95, 1.00]	0.60 [0.50, 0.69]	1.00 [0.96, 1.00]
Ekman	arc	Q2	1.00 [0.96, 1.00]	1.00 [0.96, 1.00]	1.00 [0.96, 1.00]

Table 16: The additional exact sum-convention opponent discrimination experiment. The M2 generator is compared with this exact opponent control; the other generator is the sum-convention M3. Entries give probabilities and Wilson 95% CIs. The mean-convention tests above remain part of the record.

Design / setting	Features	Gen.	Correct sign	Detected alignment	Detected organization
Helm preset	chord	M2	1.00 [0.96, 1.00]	0.91 [0.84, 0.95]	0.13 [0.08, 0.21]
Helm preset	chord	M3 sum	1.00 [0.96, 1.00]	0.64 [0.54, 0.73]	0.01 [0.00, 0.05]
Helm preset	arc	M2	0.84 [0.76, 0.90]	0.21 [0.14, 0.30]	0.03 [0.01, 0.08]
Helm preset	arc	M3 sum	1.00 [0.96, 1.00]	1.00 [0.96, 1.00]	0.00 [0.00, 0.04]
Ekman preset	chord	M2	1.00 [0.96, 1.00]	0.74 [0.65, 0.82]	0.00 [0.00, 0.04]
Ekman preset	chord	M3 sum	0.58 [0.48, 0.67]	0.02 [0.01, 0.07]	0.04 [0.02, 0.10]
Ekman preset	arc	M2	0.83 [0.74, 0.89]	0.14 [0.09, 0.22]	0.00 [0.00, 0.04]
Ekman preset	arc	M3 sum	0.53 [0.43, 0.62]	0.02 [0.01, 0.07]	0.00 [0.00, 0.04]
Helm empirical	chord	M2	1.00 [0.96, 1.00]	0.72 [0.63, 0.80]	0.36 [0.27, 0.46]
Helm empirical	chord	M3 sum	0.99 [0.95, 1.00]	0.71 [0.61, 0.79]	0.00 [0.00, 0.04]
Helm empirical	arc	M2	0.99 [0.95, 1.00]	0.39 [0.30, 0.49]	0.14 [0.09, 0.22]
Helm empirical	arc	M3 sum	0.94 [0.88, 0.97]	0.28 [0.20, 0.37]	0.00 [0.00, 0.04]
Ekman empirical	chord	M2	0.88 [0.80, 0.93]	0.14 [0.09, 0.22]	1.00 [0.96, 1.00]
Ekman empirical	chord	M3 sum	1.00 [0.96, 1.00]	0.95 [0.89, 0.98]	0.99 [0.95, 1.00]
Ekman empirical	arc	M2	0.96 [0.90, 0.98]	0.26 [0.18, 0.35]	1.00 [0.96, 1.00]
Ekman empirical	arc	M3 sum	1.00 [0.96, 1.00]	0.89 [0.81, 0.94]	1.00 [0.96, 1.00]

G Gray and Independent MLDS Results

The gray training bootstrap samples triads with replacement, then draws successes from each triad’s empirical Bernoulli distribution with the corresponding resampled count. No held-out response enters g or f . The original implementation’s string-based anti-join is not used to split repeated rows; explicit triple identifiers define the folds. Responses predict selection of test two, consistent with the reference likelihood.

Table 17: Every prespecified gray specification. Shape is p for M1 and $\log k$ for the free-log bridge; the M0 value one denotes the fixed linear shape. The ratio is Eq. 7. All scores are held out; CIs are conditional training-bootstrap intervals.

Model	NLL/trial	95% CI	Shape	95% CI	Ratio and CI
g_M0	0.64278	[0.64250, 0.64383]	1.000	[1.000, 1.000]	1.000 [1.000, 1.000]
g_M1	0.63672	[0.63625, 0.63751]	0.000	[0.000, 0.081]	0.756 [0.756, 0.772]
g_log	0.63675	[0.63622, 0.63758]	2.318	[2.181, 2.471]	0.755 [0.743, 0.767]
Lstar_M0	0.65670	[0.65621, 0.65740]	1.000	[1.000, 1.000]	1.000 [1.000, 1.000]
Lstar_M1	0.65182	[0.65171, 0.65241]	0.000	[0.000, 0.101]	0.756 [0.756, 0.776]
Y_M0	0.68963	[0.68901, 0.69062]	1.000	[1.000, 1.000]	1.000 [1.000, 1.000]
Y_M1	0.68884	[0.68787, 0.69050]	0.000	[0.000, 0.204]	0.756 [0.756, 0.806]

The near-collinear restriction retains every gray triad, up to floating-point error, and therefore leaves all table entries unchanged. This numerical check does not eliminate possible contextual curvature in the *perceived* neutral axis; the replication retains the original study’s geodesic assumption. The published gray magnitude is used only for interpretation after our own fitting, not as an objective or an optimization target.

Table 18: Independent MLDS validation, separately by context. Noise is relative to an endpoint-normalized interval scale. The gain is linear physical-luminance NLL minus fitted-scale NLL; all intervals use training-participant bootstraps.

Context	NLL/trial	95% CI	Gain and CI	Noise and CI
low transp dark	0.441	[0.433, 0.463]	0.071 [0.050, 0.079]	0.252 [0.213, 0.291]
low transp light	0.430	[0.423, 0.449]	0.048 [0.034, 0.053]	0.295 [0.251, 0.334]
plain	0.430	[0.425, 0.450]	0.078 [0.063, 0.081]	0.230 [0.192, 0.267]
transp dark	0.409	[0.402, 0.433]	0.086 [0.063, 0.093]	0.226 [0.188, 0.261]
transp light	0.414	[0.408, 0.431]	0.057 [0.044, 0.061]	0.263 [0.225, 0.298]

H Reproducibility and Deviation Record

The frozen specification is `analysis_plan.md`; `analysis_plan_addendum.md` records the exact opponent convention before its added fits. The original mean convention and all earlier results remain included. `attempts_log.md` records execution adaptations and corrections. `code/reproduce.sh` rebuilds every analysis and the PDF. Data-processing seed 20261004 and all indexed bootstrap/synthetic seeds are saved with the folds and training multiplicities. Input hashes and the frozen-plan digest are in the conversion and provenance logs.

All tables and figures derive from `logs/summary.json`; prose values derive from `logs/numbers.tex`, with a JSON value map. Full fits, out-of-fold predictions, generator truth, recovery, and resampling records are retained. There are 96,000 chromatic bootstrap fits and 76,800 synthetic diagnostic fits, all reporting optimizer convergence; gray refits also converge. Successful termination does not prove global optimality or identifiability. Multistart objectives, boundaries, bootstrap iteration counts, numerical derivative checks, fold-isolation checks, and exact software versions are documented in the saved logs and verification report. No specification was discarded for poor performance.